

Ajay Venkatesh

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EDUCATION

New York University Master of Science in Computer Engineering (GPA – 3.9/4)	Aug 2024 - Aug 2026
Visvesvaraya Technological University Bachelors in ECE (GPA - 8.9/10)	Aug 2017 - Aug 2021

SKILLS

Programming languages:, Python, Java, C++, JavaScript, SQL.
Web Development: Rest API, GraphQL API, Microservices, React, Node.js, Express.js, Sequelize.
Database: RDBMS (MySQL, PostgreSQL), NoSQL (MongoDB, JSON), Dataverse.
Cloud: AWS (EC2, S3, Amplify, Lambda Function, API Gateway, Cognito, IAM, Q, SageMaker, CloudWatch, CloudFront).
Tools: Git, Azure DevOps, Visual Studio, Jupyter, Postman, Docker, Power Automate, Power BI, QuickSight.
Methodology: Agile, Waterfall, Scrum.
AI/ML: Classification, Pattern Recognition, Data Extraction, MLOps, Neural Network, PCA, LLM.

EXPERIENCE

New York University, NYC, US	Aug 2024 - Present
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Software Developer (On-Campus)

- Contributed to an AI-driven Remote Patient Monitoring System, awarded \$100K by the **National Science Foundation, USA**, employing React Native, TypeScript, and Cognito authentication for real-time health monitoring.
- Authored **parallelized ETL pipelines** in Python to convert unstructured NoSQL data into structured SQL using GCP Cloud Functions. Leveraged concurrent processing of data chunks, reducing transformation time by 60%.
- Developed a React website to host **QuickSight dashboard** handling 50K+ records, utilized Cognito, Amplify, and Lambda functions to establish login, role-based access control, and Stripe integration for payment gateway.

Price Water House Coopers, (PwC)

Senior Software Engineer

Jul 2023 - Aug 2024

- Designed and implemented a **distributed backend system** on Microsoft Azure for a cloud-based health platform, integrating with Microsoft Dynamics for secure data flow and leveraging **Azure services** for scalability, fault tolerance, and event-driven processing.
- Led code reviews and mentored 5 interns on software design principles, test-driven development, and performance optimization, reducing production bugs by 25% and accelerating **CI/CD pipeline** efficiency in an **Agile practice**.
- Built a scalable **B2B web platform** with modular backend services to modernize a legacy user onboarding system, integrating data pipelines, email event tracking, and BI report generation, resulting in 70% improvement in automation and extensibility.
- Deployed a **cognitive neural model** via RESTful APIs to automate structured data extraction from physical cards using **OCR and NLP techniques**, reducing manual processing time by 75% and enhancing system integration with client applications.

Associate Software Engineer

Aug 2021 - Jun 2023

- Engineered **role-based access control (RBAC)** and **authorization layers**, for 5+ different roles, enforcing the principle of least privilege and preventing unauthorized access through strict session validation and privilege escalation checks.
- Optimized a data pipeline by implementing **parallel processing for cleaning, mapping, and importing** 100K-row Excel files into the database, reducing execution time from 5 hours to 1 hour and improving overall efficiency by 80%.
- Engineered a **trie-based autocomplete module** to enable fast and dynamic search across users and documents, reducing average query time by 70% and improving operational efficiency for over 10K+ client records.
- Composed client and server-side custom configurations using JavaScript, plugins, custom workflows, webhooks, Business rules yielding a 35% faster server response.

PROJECTS

Advanced Expense Tracker (Java Desktop Application) (Java, JavaFX, JDBC, SQLite, JFreeChart)

- Architected a modular Java application supporting both individual and group expenses with secure login, leveraging **OOP principles and design patterns** (MVC and DAO) for separation of concerns.
- Implemented concurrent transaction handling and real-time cost splitting using **multithreading and synchronized blocks** to ensure thread-safe database updates in group expense scenarios.
- Integrated visual analytics with JFreeChart and added CSV export using Java I/O streams, increasing data accessibility and reducing manual financial reporting time by 70%.

Hospital Management System (React, AWS, GraphQL API, DynamoDB)

- Designed a React application to manage Appointments, Patient and Doctor records, Lab Reports and Medical History.
- Modeled clinic locations as nodes in a weighted graph and applied **Dijkstra’s algorithm** to match users with the nearest available doctor based on actual travel distances, improving booking efficiency by 45%.
- Fine-tuned a transformer-based **LLM from Hugging Face** to predict patient health conditions with 88% accuracy; deployed via AWS SageMaker and integrated into the doctor dashboard for real-time, patient-specific diagnostic support.

CERTIFICATIONS

AWS Cloud Practitioner	Microsoft Developer Associate	Power BI Data Analyst Associate
Microsoft Solution Architect Expert	Azure Data Scientist Associate	Azure Data Fundamentals

PUBLICATIONS

Supervised Machine Learning System Based Segmentation and Classification of Stroke Using Deep Learning Techniques ([IEEE](#))
An Aquila-optimized SVM classifier for Diabetes prediction ([IEEE](#))

TRAININGS

Data Structures and Algorithms (NPTEL- IIT Kanpur) »	Jul 2019 - Sep 2019
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